

- 1**
- (a)** Use $v_y = u_y + a_y t$
 At max height, $v_y = 0$
 $0 = 30 \sin 60^\circ - (9.81) t$ M1
 $t = 2.6 \text{ s}$ A0
- (b)** initial velocity = 26 m s^{-1} , final velocity = -26 m s^{-1} B1
 straight line intersects x-axis at 2.6 s and ends at 5.2 s (or 5.3 s) B1
- (c)** **(i)** horizontal line at $u_x = 15 \text{ m s}^{-1}$ from $t = 0$ to 5.2 s (or 5.3 s) B1
(ii) downward sloping curve from 15 m s^{-1} and its gradient decreases numerically B1
 ending before $t = 5.20 \text{ s}$ B1
- (d)** Calculate displacements of first and second object to be 34.4 m and 5.84 m respectively using 2.6 s C1
 (or 34.5 m and 5.31 m using 2.65 s)
 Displacement between objects = 29 m A1
- 2**
- (a)** Force is the rate of change of momentum B1